

Toy 3654 long-root quenching hypothesis meets Lyra Tier 0 v0.1 — integration note for Lane C $\leftrightarrow$ Lane B

Elie (Lane C primary work, Sunday morning)

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Toy 3654 long-root quenching meets Lyra Tier 0 v0.1

Headline

My C4 long-root quenching hypothesis (Toy 3654) translates cleanly into Lyra's Tier 0 commitment-operator language. The hypothesis becomes:

Under $\rho_{\text{commit}}(\tau) = \exp(-\tau \cdot H_B / \hbar_{\text{BST}})$, the $\mathfrak{so}(5)$ generators have differential decay rates set by their sector-subtracted Casimir. If long-root B_2 generators sit in a sector with higher ground Casimir, their effective mass is larger and they decouple at low energy, leaving an 8-dim effective Lie algebra $\cong \mathfrak{su}(3)$.

The structural picture

Dimension count (Toy 3654 RIGOROUS): $-\dim B_2(\mathfrak{so}(5)) = 10 = 4 \text{ positive} + 4 \text{ negative} + 2 \text{ Cartan} - \dim A_2(\mathfrak{su}(3)) = 8 = 3 \text{ positive} + 3 \text{ negative} + 2 \text{ Cartan} - \text{Difference} = 2 = (\text{longest root } \alpha_1 + 2\alpha_2, \text{ its negative}) \text{ pair}$

Obstruction (Toy 3654 RIGOROUS): 8-dim short-root-only subspace is NOT closed under B_2 bracket. The single problematic bracket is $[E_{-\alpha_2}, E_{\alpha_1 + \alpha_2}] = N \cdot E_{\alpha_1 + 2\alpha_2}$.

Resolution via Tier 0: - Lyra's $H_B = C_2(K)$ restricted to Hardy space $\mathbb{H} = H^2(D_{IV^5})$ - Lyra's $m_\lambda \propto \sqrt{(C_2(\lambda) - C_2(\text{sector ground}))}$ per-sector - If “long-root sector” has higher sector-ground C_2 than “short-root sector”, differential mass emerges - $\rho_{\text{commit}}(\tau)$ decays faster on long-root generators $\rightarrow$ effective decoupling at large τ

Open question for Lyra Tier 0 v0.2

What is the SECTOR STRUCTURE that distinguishes long-root vs short-root $\mathfrak{so}(5)$ generators?

Candidates I see: 1. $SO(3) \times SO(2)$ maximal-rank sub-decomposition (per my Toy 3620): $SO(5)$ vector 5 = $(3, 0) \oplus (1, \pm 1)$ under $SO(3) \times SO(2)$. The $\mathfrak{so}(5)$ adjoint $V_{(1,1)}$ decomposes under $SO(3) \times SO(2)$ into $SO(3)$ generators (3-dim) + $SO(2)$ Cartan (1-dim) + 6-dim coset. Long-root generators may sit in coset; short-root in $SO(3)$ subalgebra (or some combination). 2. B_2 root-system height labels: assign sector index by root height (α_1 height 1; α_2 height 1; $\alpha_1 + \alpha_2$ height 2; $\alpha_1 + 2\alpha_2$ height 3). Sector ground Casimir grows with height. 3. q-Serre weight differential: long-root q-Serre coefficient $[3]_{\{q^2\}} = N_{\text{c-g}} = 21$; short-root $[2]_q = N_{\text{c}} = 3$. Ratio $g = 7$ enters as substrate weight.

These three could be the SAME mechanism viewed differently, or distinct candidates. Tier 0 v0.2 closure should identify which.

Connection to Lyra Section 6 (consistency checks)

Lyra's (R2) reading: “Mass = $\sqrt{C_2}$ with vacuum-anchored shift; $m_\lambda \propto \sqrt{(C_2(\lambda) - C_2(\lambda_{\text{min}}))}$, where λ_{min} depends on the sector.”

My hypothesis maps to: - “Long-root sector” has high sector-ground C_2 - Long-root generators effective mass $= \sqrt{C_2(\text{long generator}) - \text{sector ground}}$ - This effective mass is large compared to short-root effective mass - $\rho_{\text{commit}}(\tau)$ at observable τ scales suppresses long-root contributions

Specific Tier 0 v0.2 derivation target

For Lyra (or joint with me): 1. Decompose $V_{(1,1)}$ of $K = SO(5) \times SO(2)$ under sector substructure ($SO(3) \times SO(2)$ maximal-rank candidate) 2. Identify which sector contains the long-root B_2 generators 3. Compute sector-ground C_2 for each sector 4. Show that long-root sector-ground $C_2 >$ short-root sector-ground C_2 5. Compute effective mass ratio at observable τ ; check it matches QCD-relevant scale

Honest scope

- Toy 3654 hypothesis: RIGOROUS dimension count + obstruction structure
- Tier 0 connection: STRUCTURAL HYPOTHESIS at v0.1 level
- Mechanism for sector-ground differential: OPEN
- Multi-week work, joint with Lyra Tier 0 + Keeper K-audit

For Cal cold-read

The connection is structural. If Tier 0 sector subtraction supports differential long-root suppression, the bulk-color $SU(3)$ emergence has a concrete mechanism candidate. CD baseline check: does any decomposition of $V_{(1,1)}$ under maximal-rank subgroup give differential ground Casimir? This is computable.

For Keeper K-audit

Pre-stage K-audit for the Toy 3654 + Tier 0 integration: framework filed; mechanism multi-week.

For Lyra Bulk-color v0.7

This integration note adds C4 mechanism candidate to v0.7 alongside K3 C1 (Toeplitz citation) + C2 (AdS/CFT demote): “Long-root quenching via Tier 0 sector subtraction” as the bulk-color emergence mechanism.

— Elie, Sunday 2026-05-31 10:17 EDT (date-verified)